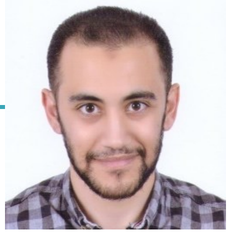
Ahmad Elshenety



Bilkent University, Turkey |+90 552 808 76 51| a.elshenety@bilkent.edu.tr | https://orcid.org/0000-0002-5783-4669

Profile

Final-year PhD candidate in Materials Science and Nanotechnology at Bilkent University, specializing in Capacitive Micromachined Ultrasonic Transducers (CMUTs). Experienced in multiphysics modeling, cleanroom microfabrication, and experimental characterization. Developed a stiction-free CMUT architecture (CMUT-FP) achieving >2.6× sensitivity with significantly reduced operating voltage. First author of 5 peer-reviewed publications. Research focus: low-voltage high-sensitivity CMUT-based transducers for wearable ultrasound.

Education

PhD in Materials Science and Nanotechnology | January 2022 - Present | Bilkent University, Turkey

- Research focus: design, simulation, microfabrication, and characterization of high-sensitivity CMUTs under the supervision of Prof. Abdullah Atalar
- CGPA: 3.87/4.0

MSc in Electrical Engineering | October 2019 | Menoufia University, Egypt

- Conducted research on PolyMUMPs-based gas sensor under the supervision of Dr. Mostafa Soliman with a thesis entitled “The parallel plate electrostatic actuator and its application as binary gas sensor”
- CGPA: 3.3/4.0

BSc in Electrical Engineering | July 2013 | Menoufia University, Egypt

- Ranking: 2nd of 180 students
- Grade: Excellence with honors
- Percentage: 88.47%

Research Experience

Bilkent University, Turkey | January 2022 - Present

- Designed and developed a novel CMUT architecture (CMUT-FP) with enhanced sensitivity and reduced voltage requirements
- Performed multiphysics simulations using ANSYS and OnScale
- Fabricated CMUT-FP devices using advanced cleanroom processes (ALD, ICP, XeF_2, etc.)
- Conducted experimental characterization using optical profilometry and electrical measurements

University of Waterloo, Canada | October 2016 - November 2016

- Completed a one-month training in the lab of Prof. Eihab Abdel-Rahman, focusing on the application of electrostatic MEMS devices as gas sensors including laser Doppler vibrometry measurements

Electronics Research Institute, Egypt | July 2015 - December 2021

- Research Assistant in the department of “Power Electronics and Energy Conversion” at the “Electronics Research Institute, Egypt”
- Participant in a national alliance for strengthening local electronics manufacturing. Worked on a gas sensor project in an interdisciplinary team, developing experience in communication between MEMS and chemistry researchers

Publications

Elshenety, Ahmad, and Mehmet Yilmaz. "Enhancing output pressure of capacitive micromachined ultrasonic transducers (CMUTs): a comparative FEM study of different pressure-boosting methods." Microsystem Technologies (2025): 1-13. https://doi.org/10.1007/s00542-025-05900-6

Elshenety, Ahmad, Merve Mintas Kucuk, and Mehmet Yilmaz. "High thickness material lift-off using multi-layer photoresist." Journal of Micromechanics and Microengineering 35, no. 2 (2025): 025012. https://doi.org/10.1088/1361-6439/adac6b

Elshenety, Ahmad, Merve Mintas Kucuk, and Mehmet Yilmaz. "Optical Characterization of Dynamic CMUTs Using Zygo Optical Profilometers: An Alternative to Laser Doppler Vibrometers." Journal of Microelectromechanical Systems (2025). https://doi.org/10.1109/JMEMS.2024.3524004

Elshenety, Ahmad, E. E. El-Kholy, Ahmed F. Abdou, Mostafa Soliman, and Mohsen M. Elhagry. "A flexible model for studying fringe field effect on parallel plate actuators." Journal of Electrical Systems and Information Technology 7 (2020): 1-14. https://doi.org/10.1186/s43067-020-00022-7

Elshenety, Ahmad, Elwy E. El-Kholy, Ahmed F. Abdou, Mostafa Soliman, Mohsen M. Elhagry, and Walaa S. Gado. "H_2S MEMS-based gas sensor." Journal of Micro/Nanolithography, MEMS, and MOEMS 18, no. 2 (2019): 025001-025001. https://doi.org/10.1117/1.JMM.18.2.025001

Skills

Microfabrication: Mask Aligner, ICP Etching, ALD, E-beam Evaporator, Sputtering, Thermal Evaporator, PECVD, XeF_2 Etching, Wet Etching

Characterization: SEM/EDX, Zygo Optical Profilometer (Dynamic/Static), Impedance Analyzer, Semiconductor Parameter Analyzer (SPA), Bruker Dektak, Keyence VKX Microscope, Probe Station, Dicing Saw, LPKF PCB Milling Machine

Design & Modeling: OnScale, ANSYS, L-Edit (Layout), MATLAB, Field II (basic)

Mentorship & Teaching

Teaching Assistant (Introduction to Micro and Nanofabrication) | Bilkent University | January 2026 – May 2026

- Delivered hands-on cleanroom training to graduate students on the operation of advanced microfabrication and characterization tools, including Zygo Optical Profilometer, Mask Aligner, ICP Etcher, Plasma Asher, Thermal Evaporator, Sputtering, ALD, PECVD, and E-beam Evaporator
- Supervised the students through the microfabrication of their course projects in cleanroom

Undergraduate Research Mentor | Bilkent University | August 2025 – September 2025

- Provided technical training on the LPKF PCB milling machine
- Guided the student through the layout and milling of a simple Bias-T circuit for CMUT excitation

Teaching Assistant (Physics Lab) | Bilkent University | January 2022 – May 2022 & January 2024 – May 2024

- Facilitated laboratory instruction and safety for undergraduate experiments
- Evaluated technical reports and provided feedback on data analysis and experimental methodology
- Advised students on the selection, design, and execution of individual course projects
- Assessed final project outcomes for 60+ students per semester

Conferences & Workshops

International Workshop on Micromachined Ultrasonic Transducers (MUT) | France (June 2025)

- Oral Presentation: [Fabrication of Capacitive Micromachined Ultrasonic Transducer with a Force Plate (CMUT-FP)]

19^{th} Nanoscience and Nanotechnology Conference (NanoTR) | Turkey (August 2025)

- Poster Presentation: [Capacitive Micromachined Ultrasonic Transducers with Integrated Force Plates (CMUT-FP): Simulation and Fabrication]

Peer Review

December 2024: IEEE Sensors Journal

Languages

English: Excellent (TOEFL iBT: 93/120)
Arabic: Native

References

Prof. Abdullah Atalar
Position: Professor at Bilkent University, Turkey
Work Address: Üniversiteler, 06800 Çankaya/Ankara, Turkey
Email address: aatalar@bilkent.edu.tr

Prof. Ayhan Bozkurt
Position: Associate Professor at Sabanci University, Turkey
Work Address: Sabancı University, Orta Mah. Üniversite cd. No:27, Tuzla/Istanbul, Turkey
Email address: abozkurt@sabanciuniv.edu